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Published in Towards Data Science

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May 28, 2020 · 10 min read · ✨ · Listen

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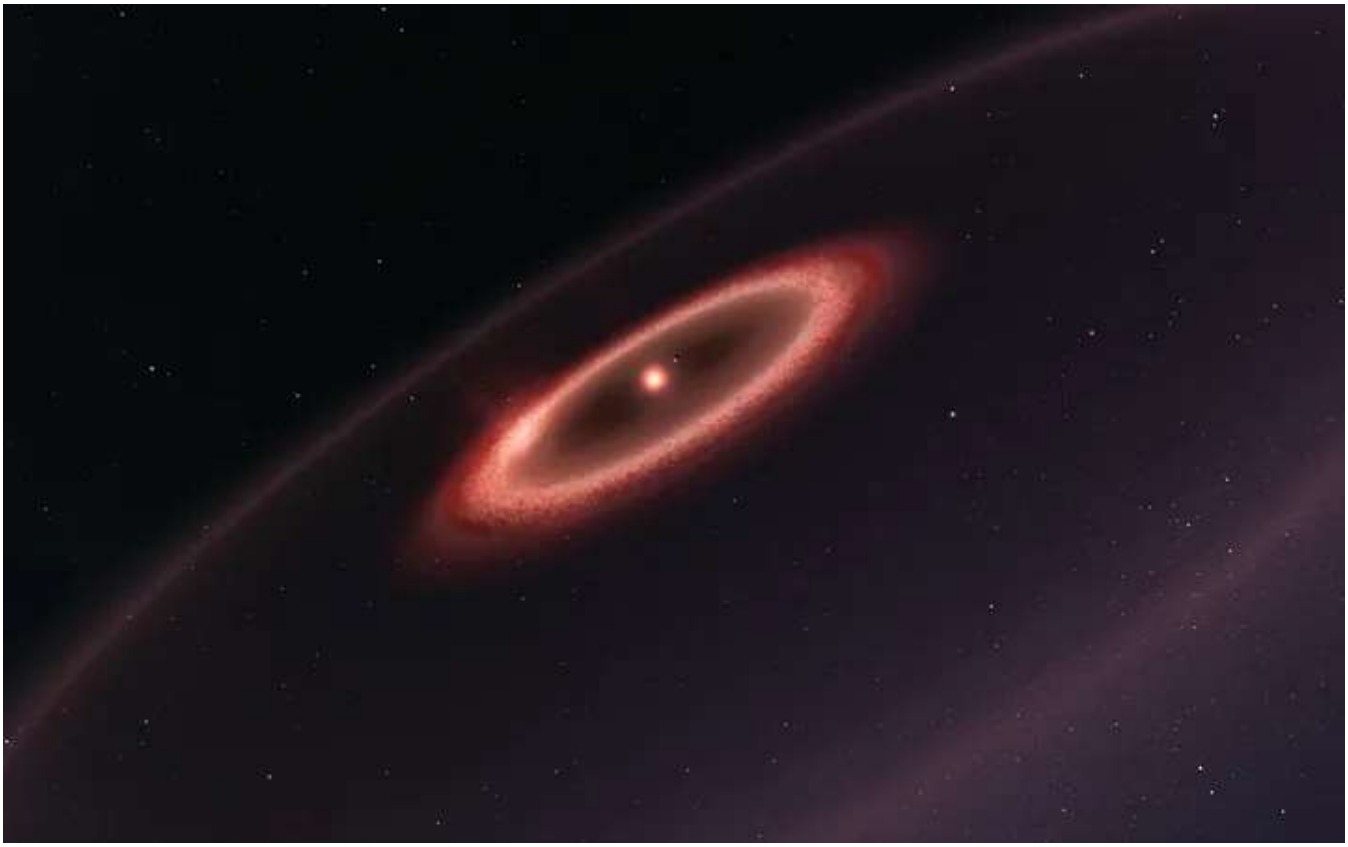
SPACE SCIENCE WITH PYTHON

Space Science with Python — Did we observe everything?

Part 11 of the tutorial series is about a Data Scientist's and Space Scientist's nightmare: Statistical Bias!



Artist's impression of a dust and debris disk or belt around our Neighbour star Proxima Centauri. Our Solar System has the main belt, Kuiper belt and th Oort Cloud. Did we observe all Oort Cloud comets? Credit: ESO/M. Kornmesser; License: Cre ribution 4.0 International License

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Preface

This is the 11th part of my Python tutorial series "Space Science with Python". All codes that are shown here are uploaded on GitHub. Enjoy!

This tutorial's database has been created in a previous session, but it is also uploaded on my GitHub repository.

Introduction

What is the average age of people living in Germany? To answer this question I go the my home town's university and survey the age distribution of all students. Afterwards I compute the average and I am done. The answer and conclusion: 23 years; Germany is a surprisingly young country!

Official numbers however say that the average age is above 40 years … *what happened?*

Probably you are suspicious of my data acquisition method and analysis. I wanted to determine the average age of all German citizens. Instead of analysing large, randomised groups or acquiring official data of all inhabitants, I *distorted* my results by choosing a completely improper surveying method. Of course, the age distribution of young students at the university does not represent the nationwide age distribution at all! I performed a so called *Selection Bias*.

Biases are present everywhere when you acquire data: due to biased observations, biased instruments (either due to their limitations or calibration), or by using wrong statistical methods (e.g., using the standard deviation for a non-symmetric distributions).

The data scientific challenge is to *unbias* the data and to derive feasible statistical results and insights. *Unbiasing* is also needed in machine learning, e.g., when a classifier requires a uniform distribution of different classes for training and validation.

But do we have a *Selection Bias* in our comet database? Let's take a look at bias effects in astronomy and space science by doing a step-by-step data exploration journey!

Malmquist Bias

How many stars do you see? And what kind of stars? Photo by Jeremy Thomas on Unsplash

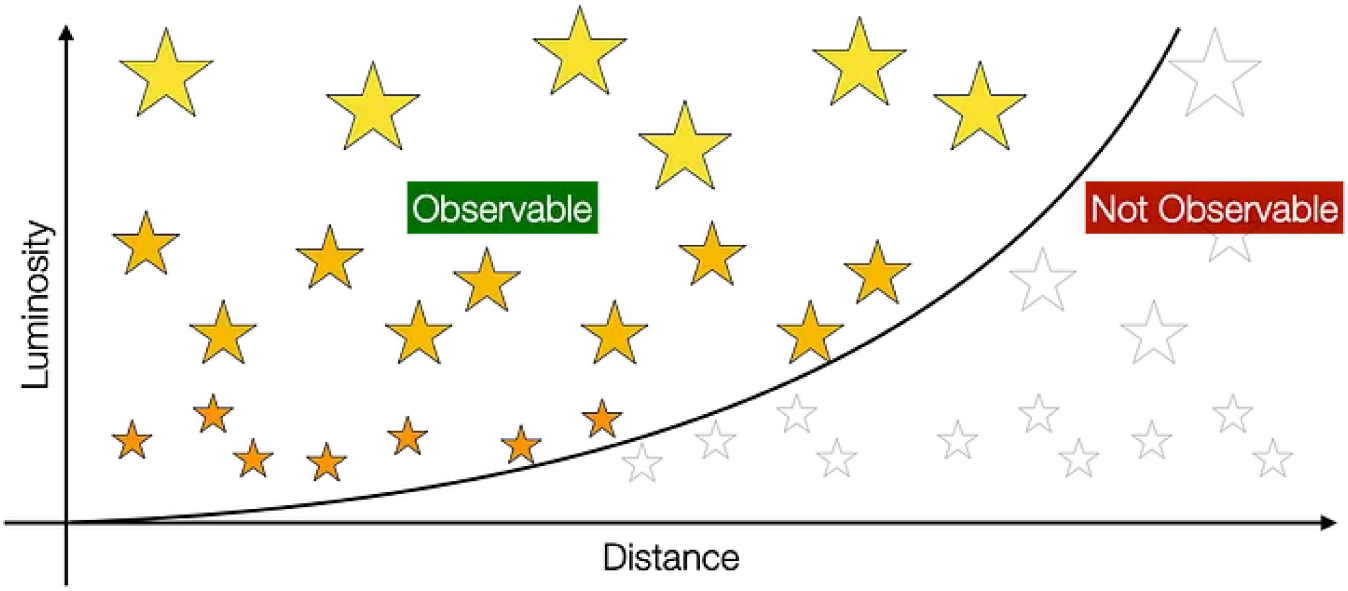
Look at the night sky. *What do you see?* The planets, our Milky Way, sometimes a falling star and of course actual stars. The stars are located at different distances. For example, our closest neighbour Proxima Centauri is around 4.2 light-years away.

The apparent brightness of the stars (so the brightness as seen from Earth) depends on the distance as well as on the intrinsic luminosity of the star itself. This intrinsic brightness is called the absolute magnitude and describes the luminosity of a star from a distance of 10 parsecs (around 32.5 light-years). So, the absolute magnitude normalises the brightness of a star with a certain distance and allows astronomers to compare the stars on a luminosity scale. There are smaller, fainter as well as larger and brighter stars in our cosmic vicinity.

Reconsidering the distance between us and the stars implies that stars with a intrinsicly fainter luminosity are less detectable at a certain distance than brighter

ones. This causes an observational bias that could lead to the wrong conclusion that there are more brighter objects in the universe.

The Swedish astronomer Gunnar Malmquist described this idea first in 1922 [1]. Observational results must be unbiased to prevent this over-estimation of brighter objects.



Schematic visualisation of the Malmquist Bias. The sketch shows the intrinsic luminosity of stars vs. their distance to an observer. A larger distance results in a worse detectability of fainter stars. An observational bias results in “Observable” and “Not Observable” objects. Credit: T. Albin

Does this also apply for Solar System bodies, like comets? If so, what are the key parameters of this bias? Let’s find out.

Is our comet dataset biased?

Our today’s lesson will use already knwon modules. Later, we need the package *scikit-learn* that can be easily installed via *pip*:

```
pip3 install scikit-learn
```

Do not worry, we will start easily with this mighty data science package. For now, we will only need a simple linear regression model that helps us to reveal some important insights.

But first things first …

Part 1/9

So what kind of bias in our comet data set could correspond to the Malmquist Bias for stellar objects? Malmquist described and compared the stars with an intrinsic luminosity, the absolute magnitude. *Is there an equivalent for comets?*

You may have noticed that our comet database has indeed a column with the required absolute magnitudes. However, these values are differently defined for comets than for stars! To compute the apparent magnitude (brightness) of a comet an equation is used that depends on the activity of the comet (outgassing, dust production), the distance to our home planet as well as the distance to the Sun.

Further, the phase angle between the Sun and the comet as seen from Earth needs to be considered. The absolute magnitude is simply an offset parameter that is related to the size of the comet (*). Assuming that all comets have the same reflectivity the absolute magnitude can be used for size comparison. Important: size and absolute magnitude are negatively correlated; a larger number corresponds to a smaller comet and vice versa.

We extract the absolute magnitude (for later purposes also the perihelion) from the database (line 3) in line 7 to 9 with the *read_sql* command of *pandas*.

Part 2/9

Doing science means, to explore data: So let's get a feeling first for the data by printing some descriptive statistics and creating some data exploration plots.

Part 3/9

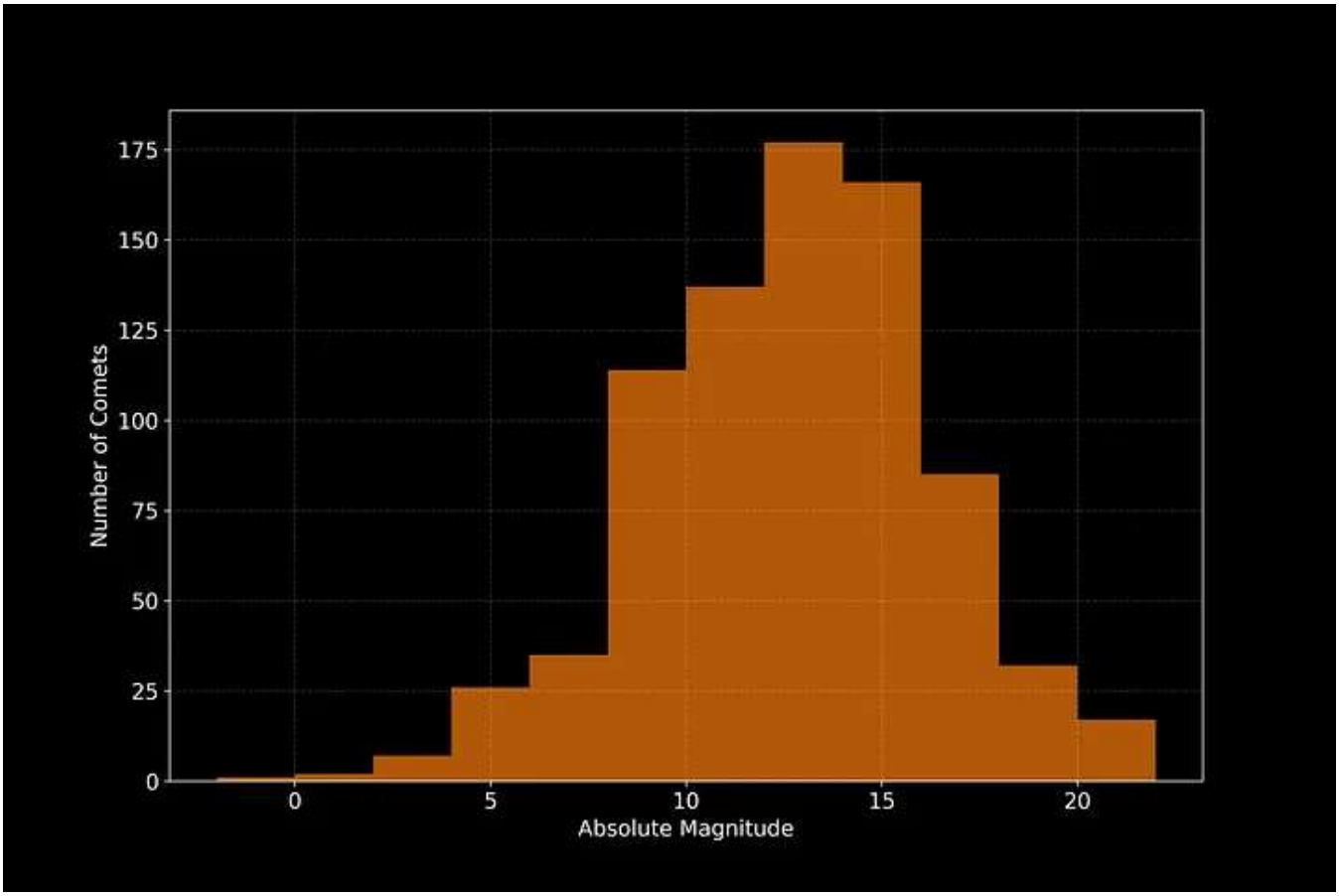
We have 799 Sun-bound comets in our Solar System (the ones we know!). The median absolute magnitude is at around 13 mag and the magnitudes vary between -2 and around 21 (*).

```
Descriptive Statistics of the Absolute Magnitude of Comets
count    799.000000
mean      12.414894
std        3.597028
min       -2.000000
```

```
25%       10.000000
50%       12.900000
75%       15.000000
max       21.200000
Name: ABSOLUTE_MAGNITUDE, dtype: float64
```

We create a simple histogram to visualise the distribution. *Maybe we can already see some bias effects?* In some tutorial we have been using the Kernel Density Estimators (KDEs) to create smooth distributions. Here, let's keep it simple by using histogram plots with a defined bin width of 2 mag (line 2). The code afterwards creates a figure (line 11) and the histogram itself (line 14 and 15). Some labels are set (line 18 and 19), a grid is activated for better readability (line 22) and the resulting figure is saved in line 25.

Let's take a look at the resulting distribution that is shown below. We see the number of comets in a histogram with bin widths of 2 mag. For now, we have observed only a few comets with a small absolute magnitude respectively large size. With larger absolute magnitude the number of known comets increases up to around 12 to 14 mag and decreases afterwards. *Are there less smaller comets in our Solar System? Or is the observation of smaller objects more difficult, causing an observational bias?*

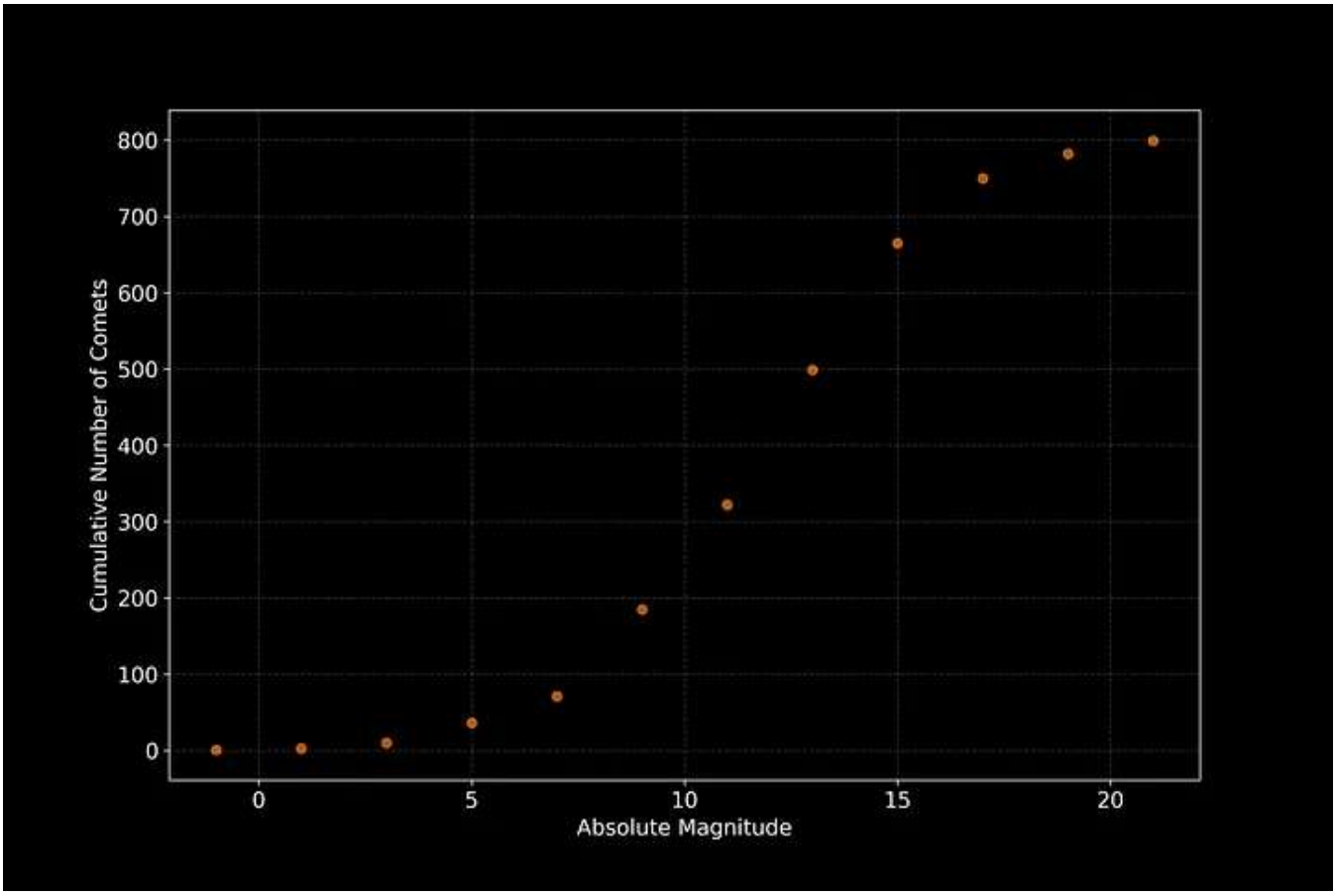


Number of comets vs. the absolute magnitude. Credit: T. Albin

Let's re-arrange the shown plot. We compute the cumulative distribution of the absolute magnitudes. We use the *numpy histogram* function to compute the histogram as shown above (line 9 and 10) and apply the function *cumsum* to calculate the cumulative distribution. Instead of creating a bar chart we visualise the results in a scatter plot. That way, possible trends are easier to determine (line 18 and 19). Please note, that the array *BINS_EDGE* contains only the positions of the bars' edges. Thus, we add a half bin width (1 mag) to the array in line 18 for the scatter dots that shall be placed between the edges.

Part 5/9

Well … the shape of the curve reminds a data scientist of the logistic regression?! But somehow this does not help, does it?

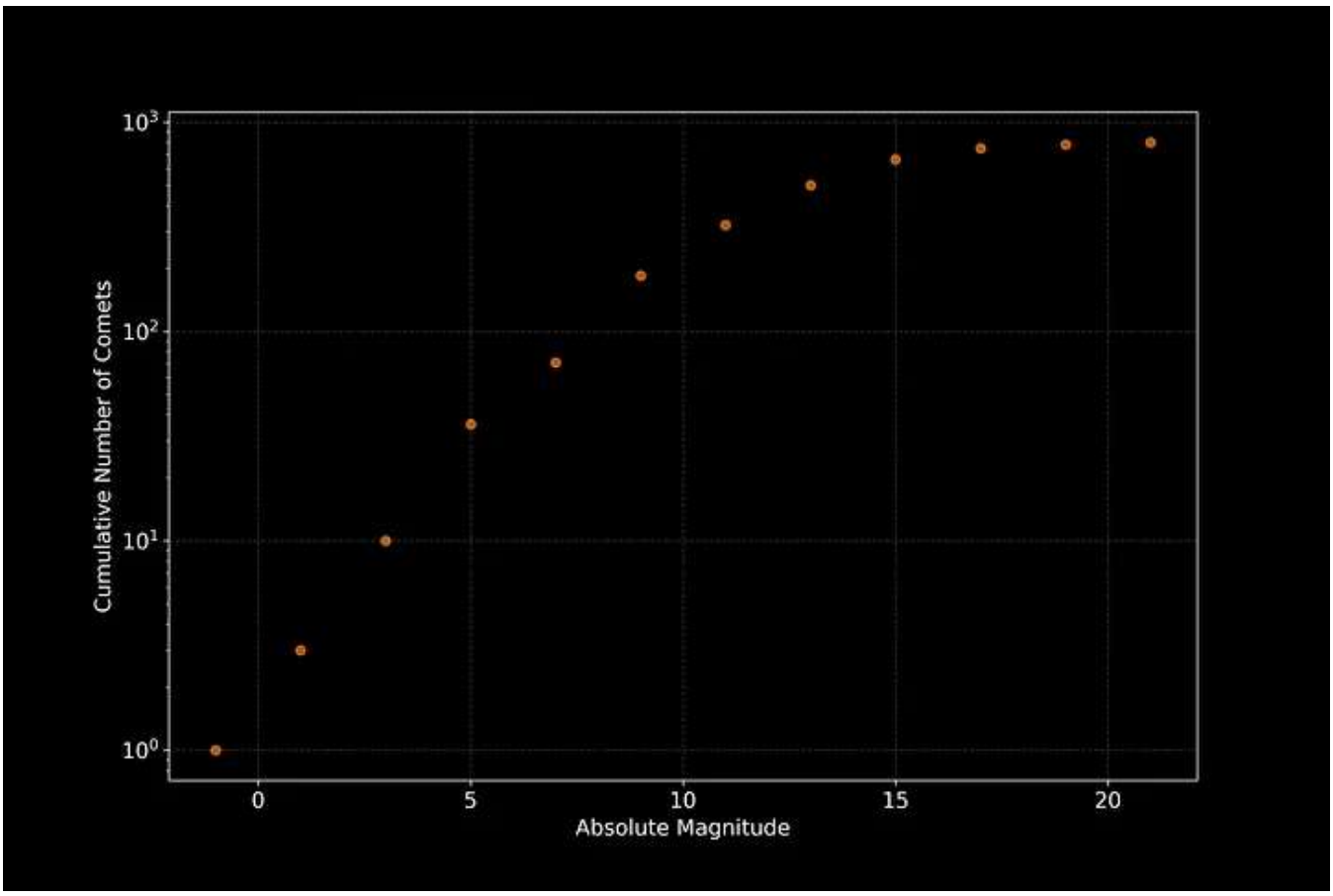


Cumulative distribution of the absolute magnitude as a scatter plot. Credit: T. Albin

Small changes, also in data exploration as we perform it today, may cause a huge impact! So let's take the same code again and add one simple line of code. Line 18 scales the y axis logarithmically and …

Part 6/9

… the resulting plot looks like this. You can see that the cumulative distribution follows a power law for smaller absolute magnitudes. At around 8 mag the power law "bends" and the cumulative distribution approaches its final number of 799 comets asymptotically.



Cumulative distribution of the absolute magnitude as a scatter plot (y axis scaled logarithmically). Credit: T. Albin

Does it help us? Let's think again about possible observational biases. Large comets have a smaller absolute magnitude. Their apparent brightness is higher as seen from Earth; compared to a smaller comet at the same distance. Consequently, similar as Malmquist's idea, it is more difficult to detect smaller comets. *But are there maybe less smaller comets in our Solar System?*

Probably not. Whether its comets, asteroids, meteoroids or so called Near-Earth Objects. The population of macroscopic objects in our cosmic vicinity follows a certain power law. The result: there are way more smaller than larger objects in our Solar System. Either by sophisticated unbiasing of observational data, or by numerical simulations one can derive this scientific insights [2, 3].

So is the gap "large" between known and unknown comets? We can easily generate a *quick-and-dirty* insight.

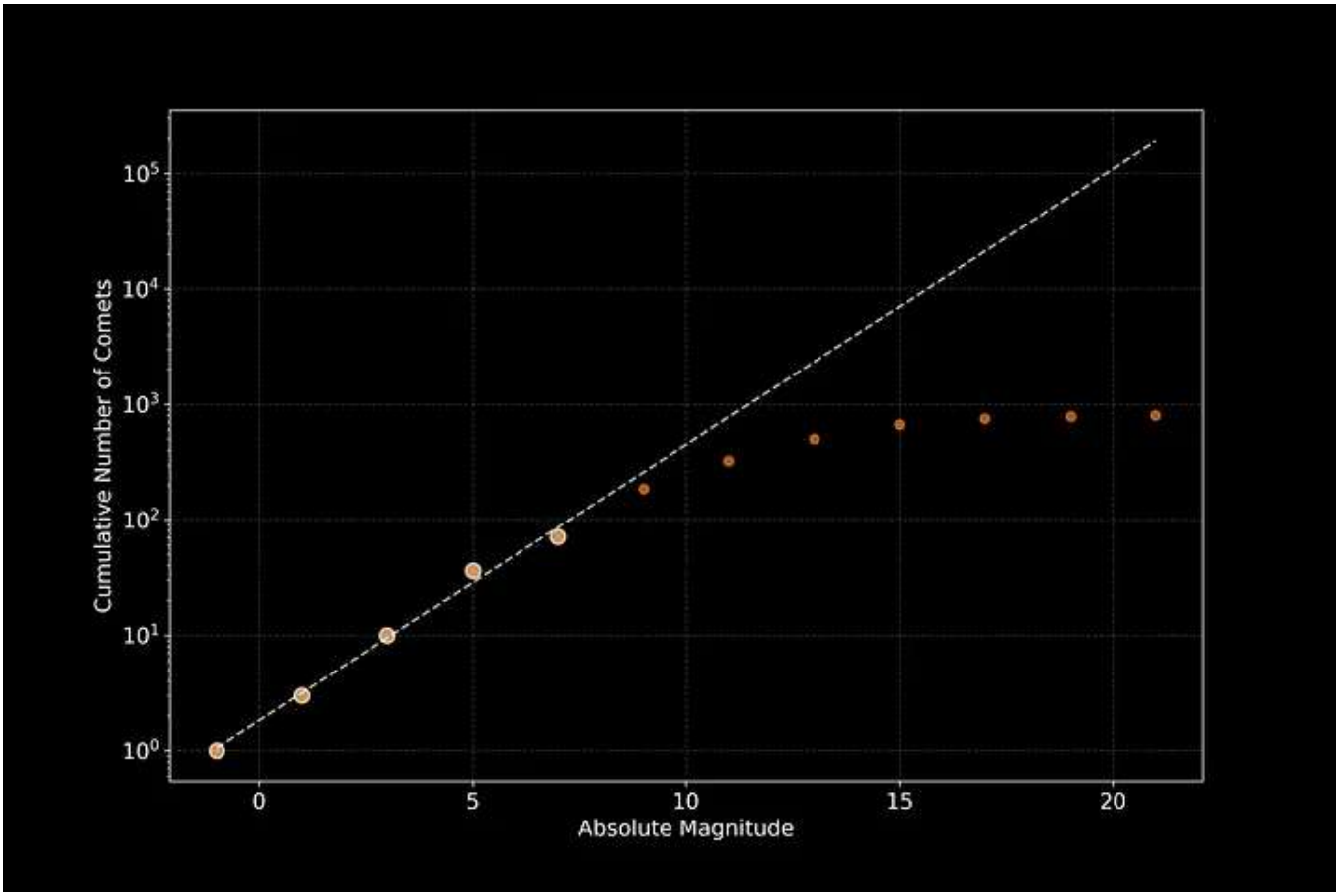
Science is often about assumptions and the idealisation of experiments or theoretical concepts. Let's assume that we know all (or a significant amount of) larger comets, let's say up to 7.5 mag. We state that the distribution applies only for the inner Solar System, since we cannot observe large comets that are e.g., in the Oort Cloud. We take the first 5 data points from the cumulative distribution (line 7 and line 11) and compute a simple linear regression model in the logarithmic domain. For this purpose we import the class *LinearRegression* from the *scikit-learn* library (line 14 and 15). We fit the model in line 18 and compute the results for the entire absolute magnitude domain in line 21.

Part 7/9

Now, we can plot the resulting linear model. Line 2 creates the figure. Line 5 and 6 plot the data points that have been used for the fitting in a different colour than the entire data set in line 9 and 10. Line 13 adds the linear regression plot and afterwards our standard formatting is applied.

Part 8/9

The figure shows a large discrepancy between our simple power law expectations (white dashed line) and the comets with a larger absolute magnitude. According to our simple law, 10^5 comets are populating our Solar System up to a cumulative magnitude of 20 mag. However, currently we only know 800 (less than a percent).



Cumulative distribution of the absolute magnitude as a scatter plot (y axis scaled logarithmically). The white dashed line is a linear fit of the first 5 logarithmically scaled data points. Credit: T. Albin

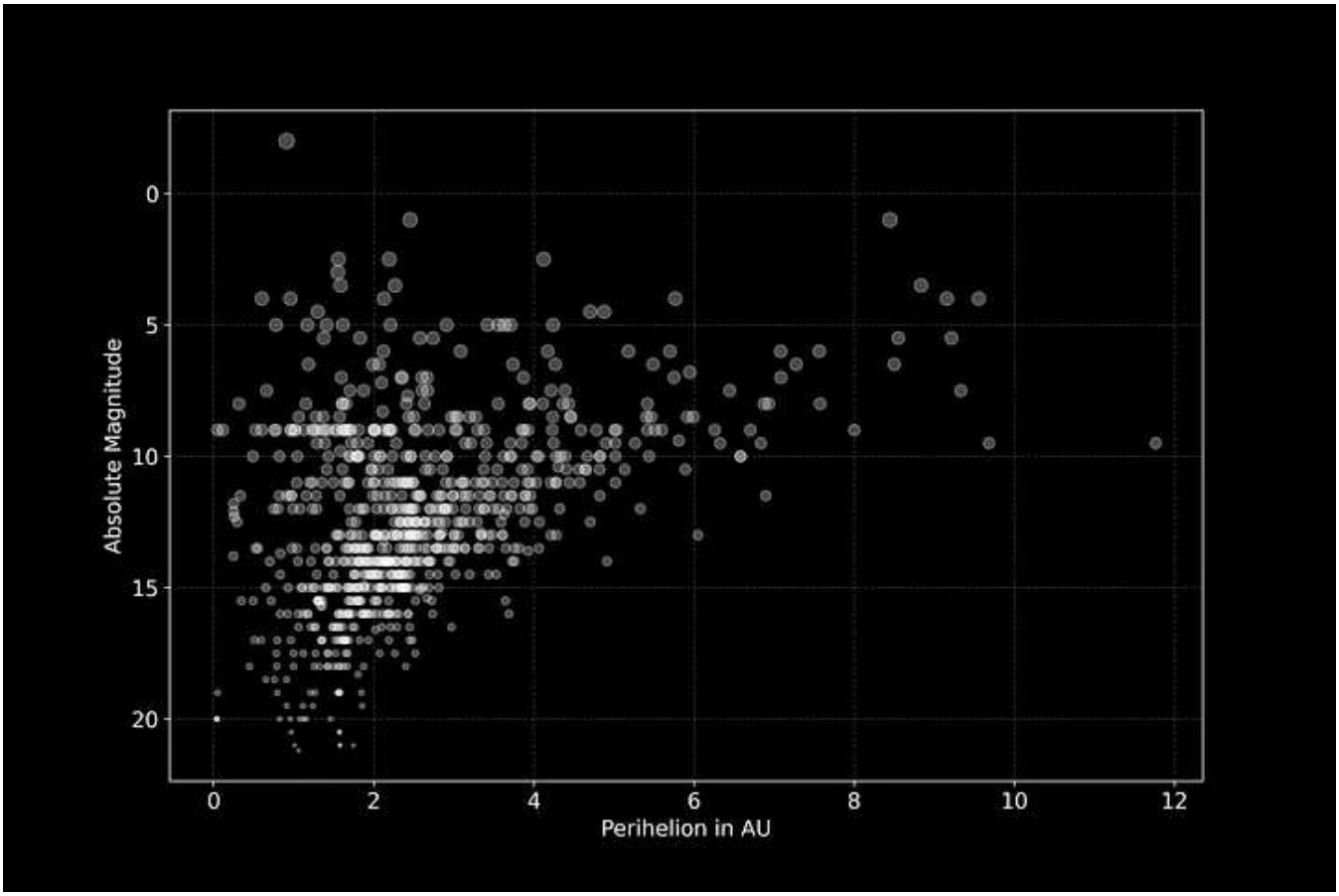
Can we explain this bias? Well first of all, we survey the night sky since a few decades. Observational technologies (optics and cameras) are improving over time. Furthermore, C type comets have a large orbital period. Some of them did not visit the inner Solar System yet and did no reach us in an observable range.

However, larger comets that do not enter the inner part of the Solar System should be easier to detect than smaller ones. *Could we "see" a Malmquist Bias analogy for comets?* Let's create a scatter plot where we plot the absolute magnitude vs. the perihelion of a comet (this could correspond to the Malmquist Bias plot shown above: luminosity vs. distance).

We create a plot (line 3) with the data points in a scatter cloud. Further, the scatter dots shall scale, depending on the absolute magnitude. A large absolute magnitude however should correspond to a smaller marker size. This is done in line 9 to 14. First, the absolute magnitudes are subtracted with the largest value (-1, otherwise the maximum value becomes 0). Then, we normalise the values and scale them (line 13, respectively 14). These marker sizes are used in the scatter plot in line 18 and 19. We invert the y axis in line 23. That way, the larger comets are in the upper part of the plot.

Part 9/9

We can see the scatter cloud from almost 0 AU to 12 AU and the absolute magnitude varies from 20 mag to around 0 mag. We can clearly see that smaller comets (respectively a higher absolute magnitude) correspond to a smaller perihelion than larger comets. Small comets need to enter the inner parts of the Solar System to become detectable for us. Reconsider the Malmquist Bias figure shown at the beginning. The shown scatter plot reminds one of this observational bias.



Scatter plot of the absolute magnitude of comets vs. the perihelion in AU. A larger absolute magnitude corresponds to a smaller comet. Varying marker sizes indicate this (not in scale). Credit: T. Albin

Conclusion

There are a lot of comets out there! Millions, maybe billions if you consider the Oort Cloud. Observational limitations cause a bias that could lead to wrong conclusions about the comet population of our Solar System. Numerical simulations are a key element in astronomy and Space Science and are needed to reveal the true nature of our cosmic vicinity.

Be aware, whether in Space Science, or any other scientific or data scientific field: Bias is everywhere! Explore, describe and understand it. Otherwise one can get into statistical pitfalls that lead to wrong conclusions.

Outlook

For now, we worked with the entire comet data set. The next tutorial sessions will focus on one particular comet: 67P/Churyumov–Gerasimenko. First, we will have a

3 D look at the comet and afterwards, we will move back to SPICE and work on a deep dive numerical approach.

Thomas

References

[1] Malmquist, K. G. (1922), *On some relations in stellar statistics*, Meddelanden fran Lunds Astronomiska Observatorium Series I, Vol. 100, p. 1–52, Bibcode: 1922MeLuF.100....1M

[2] William F. Bottke, Alessandro Morbidelli, Robert Jedicke, Jean-Marc Petit, Harold F. Levison, Patrick Michel, Travis S. Metcalfe, *Debiased Orbital and Absolute Magnitude Distribution of the Near-Earth Objects*, Icarus, Volume 156, Issue 2, 2002, Pages 399–433, ISSN 0019–1035, https://doi.org/10.1006/icar.2001.6788.

[3] James M. Bauer, Tommy Grav, Yanga R. Fernández, A. K. Mainzer, Emily A. Kramer, Joseph R. Masiero, Timothy Spahr, C. R. Nugent, Rachel A. Stevenson, Karen J. Meech, *Debiasing the NEOWISE Cryogenic Mission Comet Populations*, 2017, The Astronomical Journal, Volume 154, Number 2.

Footnotes

(*) The magnitude scale in astronomy is commonly used and an additional tutorial that explains the basics will explain it in detail.

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